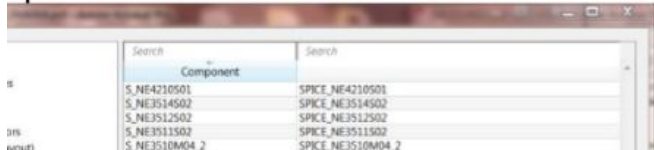


Microwave Final Design Review

12GHz 2-stage LNA

James Jagielski (jj3552)

Operating Frequency	11.4-12.7G (Starlink Customer Terminal from Satellite)
Noise Figure	<1 dB (as low as possible)
Insertion Gain (S_{21})	>20 dB (requires 2-stage design)
Output Reflection (S_{22})	<-14 dB
Input Reflection (S_{11})	<-10 dB
Board for hybrid Integration	Duroid 5880 (thickness 10 or 20 mil, ½ oz. Cu)
Active Device to be used	Any of the GaAs FET (non-linear model) from Renesas ADS library, HP ATF36077, Fujitsu HEMT's
Other requirements	Unconditionally stable 0.1-20 GHz
Other parameters to simulate	S-parameters (0.1-20 GHz), stability factor(s) Total DC power, Input 1 dB compression point (P_{1dB}), Input third-order intercept point (IIP3) ($f_1 - f_2 = 1$ MHz)



Characterizing The Device

Stability and Biasing

Component Selection - ATF-36077

<https://mm.digikey.com/Volume0/opasdata/d220001/medias/docus/803/ATF-36077.pdf>

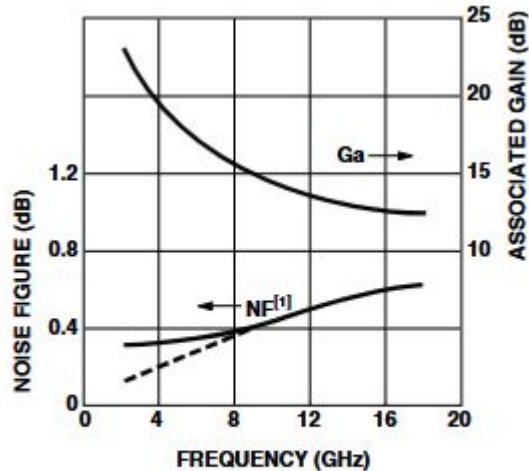
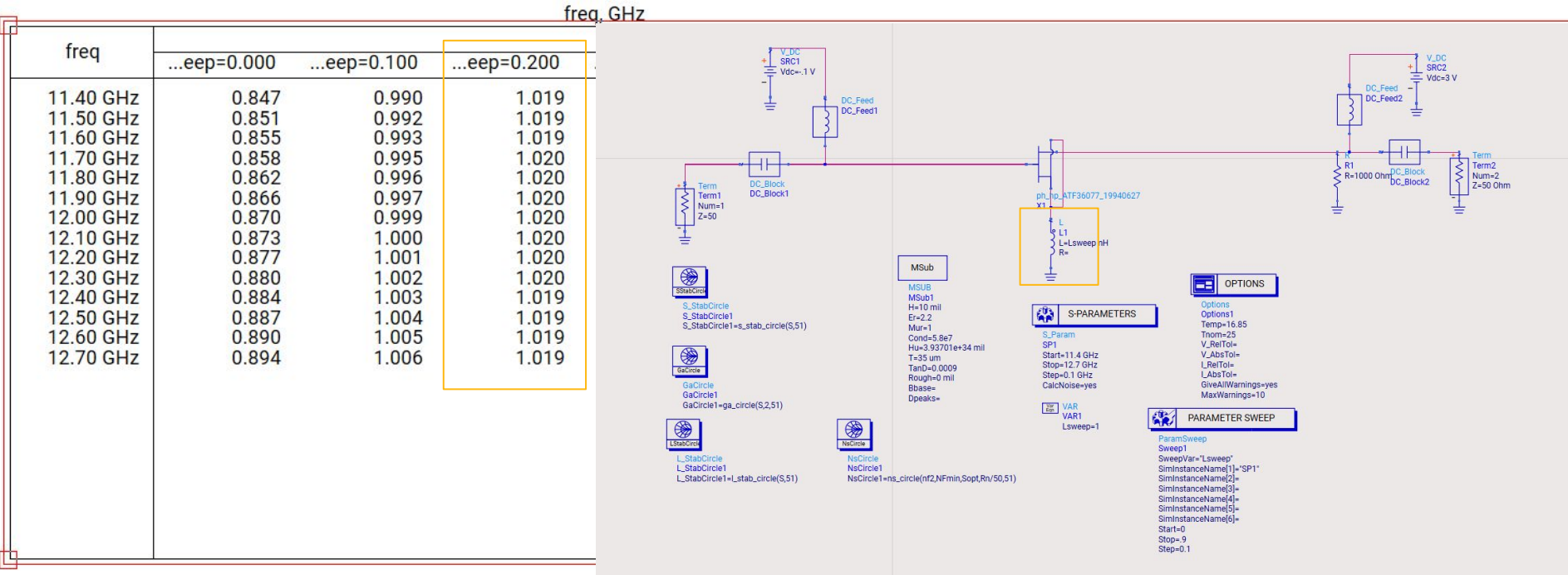


Figure 1. ATF-36077 Optimum Noise Figure and Associated Gain vs. Frequency for $V_{DS} = 1.5$ V, $I_D = 10$ mA.

- I went with the ATF-36077 for the low noise ~ 0.5 dB at 12 GHz and high gain of ~ 13 dB at 12 GHz.
- I considered alternatives for the second stage to get higher gain; however, the ATF-36077 was able to achieve the requirements easily.
 - Other alternatives would raise the total noise figure over the requirements to get the higher gain.
 - This could've been optimized, but the 36077 was reasonably close, so I ultimately kept using it.

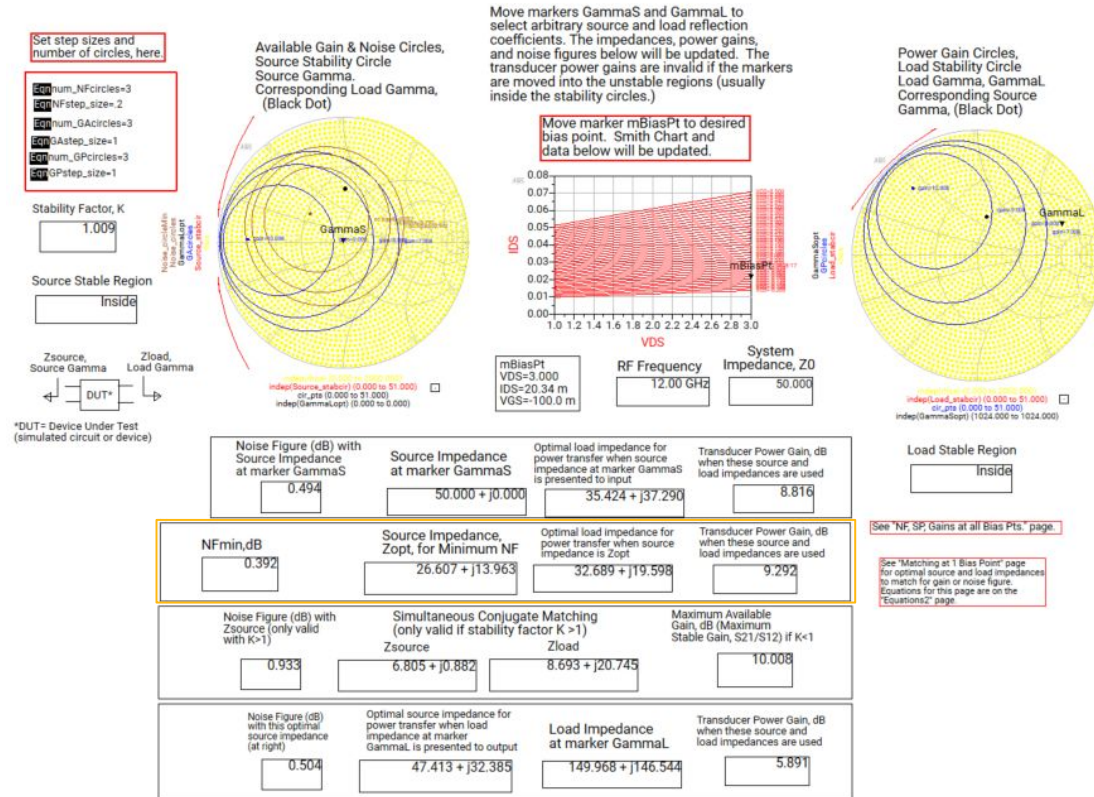
Stability Analysis

I used a bias of $-1V$ as I swept the inductor for source degeneration. Also used was a resistance of $R = 1K \text{ Ohm}$ on the drain of the device. The ideal inductance for stability was $L = .2nH$.



Bias Point

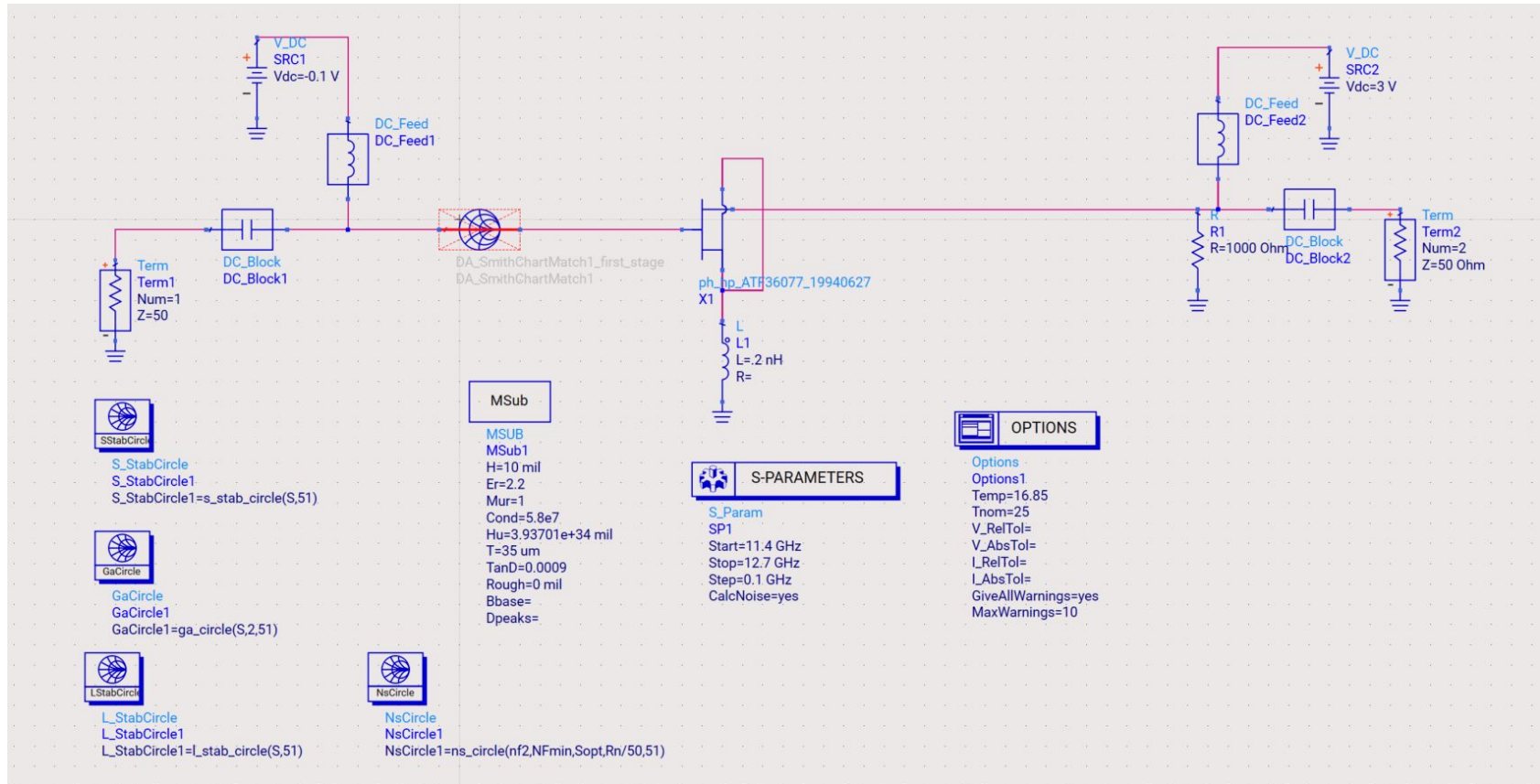
- The best bias for the gate was found to be **Vgs = -1V**.
- This gave excellent NFmin and decent enough gain.



First and Second Stage Design Walkthrough

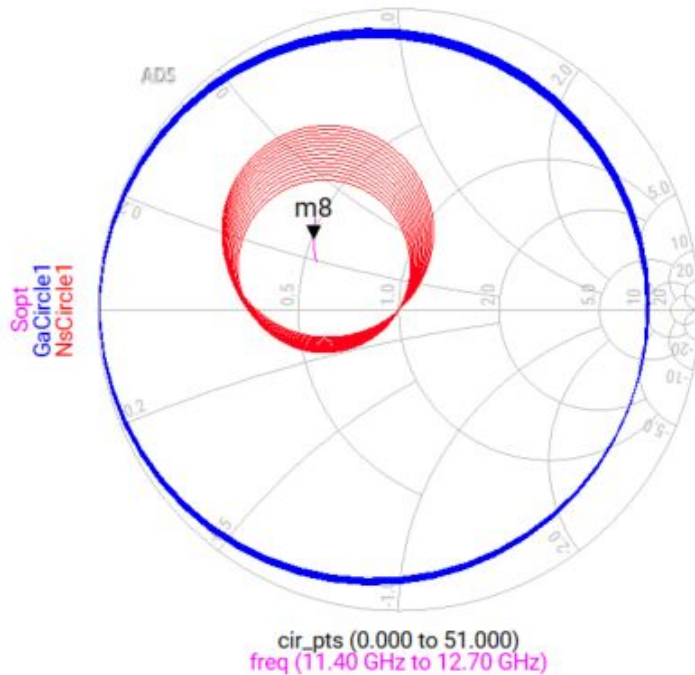
First stage is optimized for lowest noise
Second stage is optimized for Gain

First method of finding the source reflection coefficient



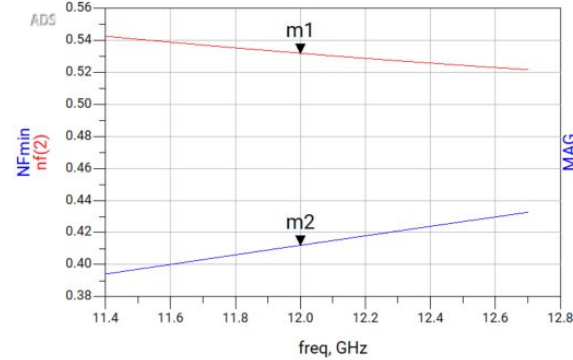
Input matching to the Sopt impedance, which is the center of the noise circle. This will provide poor gain, but great noise figure in hopes to minimize the noise in total. Results with no matching or biasing, using schematic from previous slide

m8
freq=12.00 GHz
Sopt=0.368 / 140.060
impedance = $Z_0 * (0.508 + j0.278)$



m1
freq=12.00 GHz
nf(2)=0.532

m2
freq=12.00 GHz
NFmin=0.412



$$0 * \log((\text{mag}(S_{21})^{**2}) * (1 - \text{mag}(\text{Sopt})^{**2}) / ((\text{mag}(1 - S_{11} * \text{Sopt})^{**2}) * (1 - \text{mag}(G_{\text{out}})^{**2})))$$

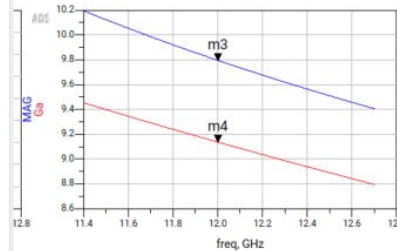
$$= (S_{22} + S_{12} * S_{21} * \text{Sopt} / (1 - S_{11} * \text{Sopt}))$$

onj(G_out)

= Sopt

m3
freq=12.00 GHz
MAG=9.795

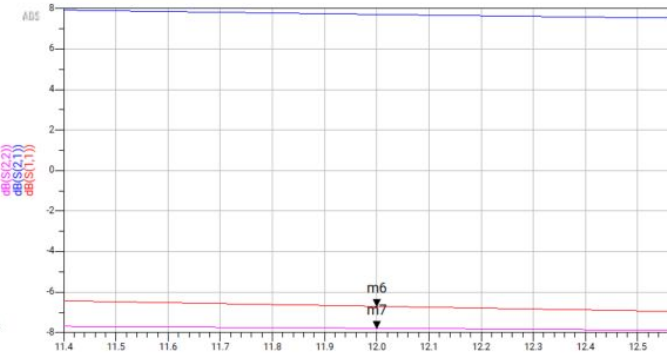
m4
freq=12.00 GHz
Ga=9.137



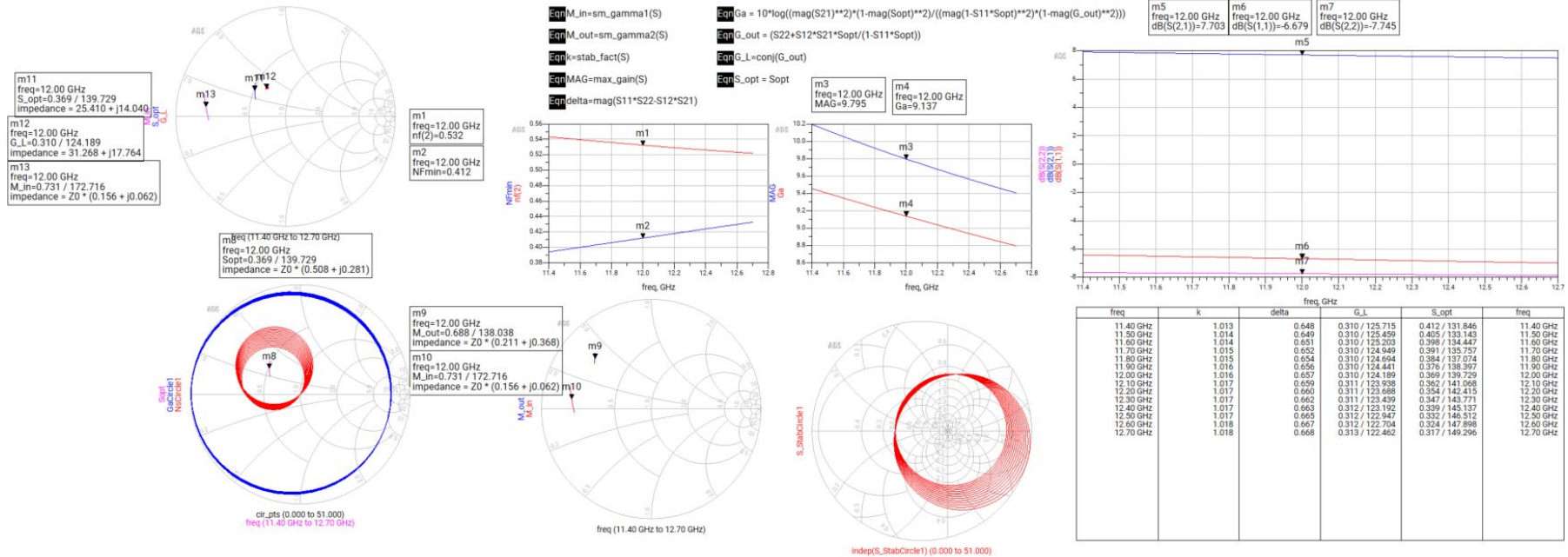
m5
freq=12.70 GHz
dB(S(2,1))=7.489

m6
freq=12.00 GHz
dB(S(1,1))=-6.700

m7
freq=12.00 GHz
dB(S(2,2))=-7.783



Total results (continuation of the previous slide)

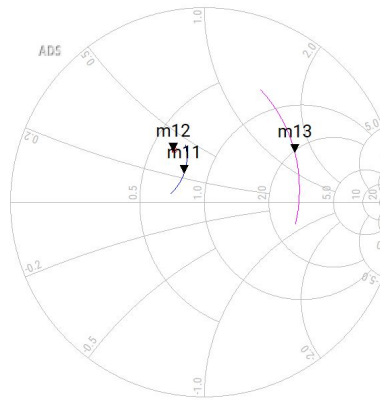


m11
freq=12.00 GHz
S_opt=0.182 / 125.129
impedance = 38.913 + j11.973

m12
freq=12.00 GHz
G_L=0.307 / 121.635
impedance = 31.986 + j18.448

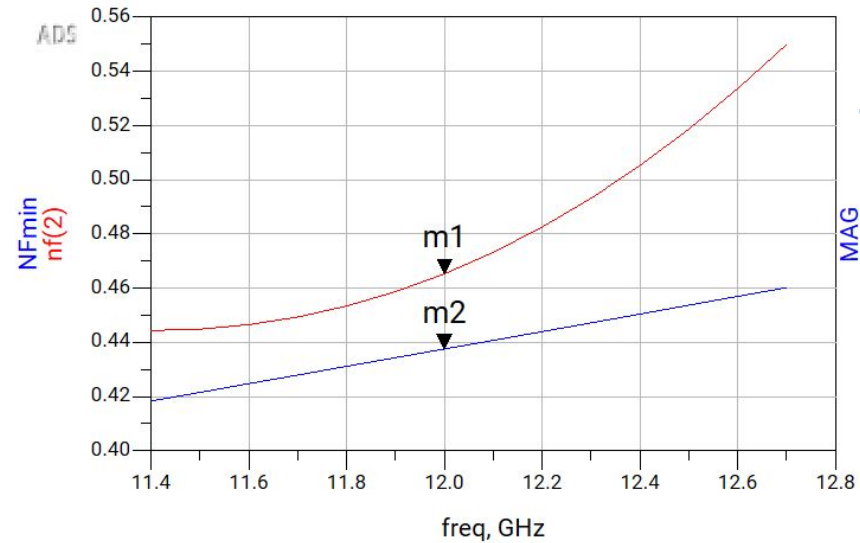
m13
freq=12.00 GHz
M_in=0.534 / 28.828
impedance = 102.241 + j73.536

M_in
S_opt
G_L

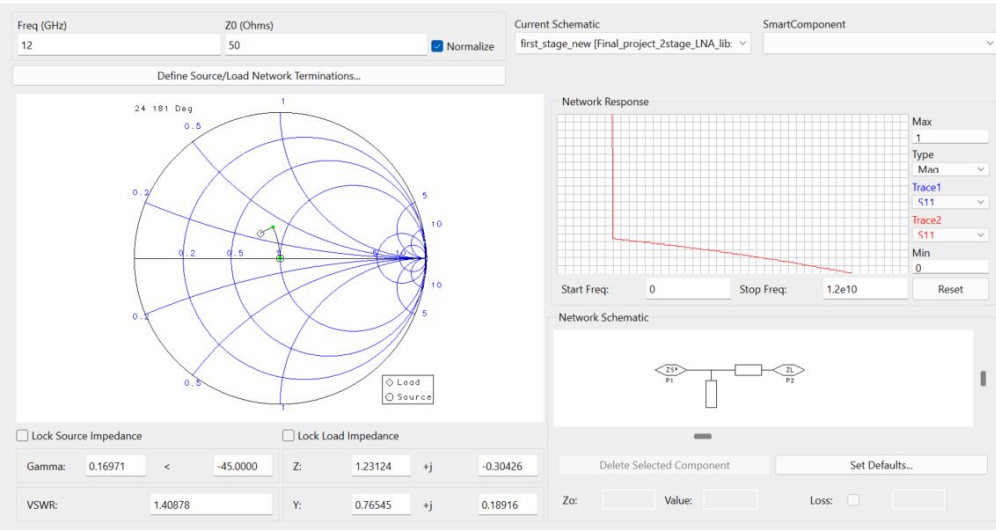


m1
freq=12.00 GHz
nf(2)=0.465

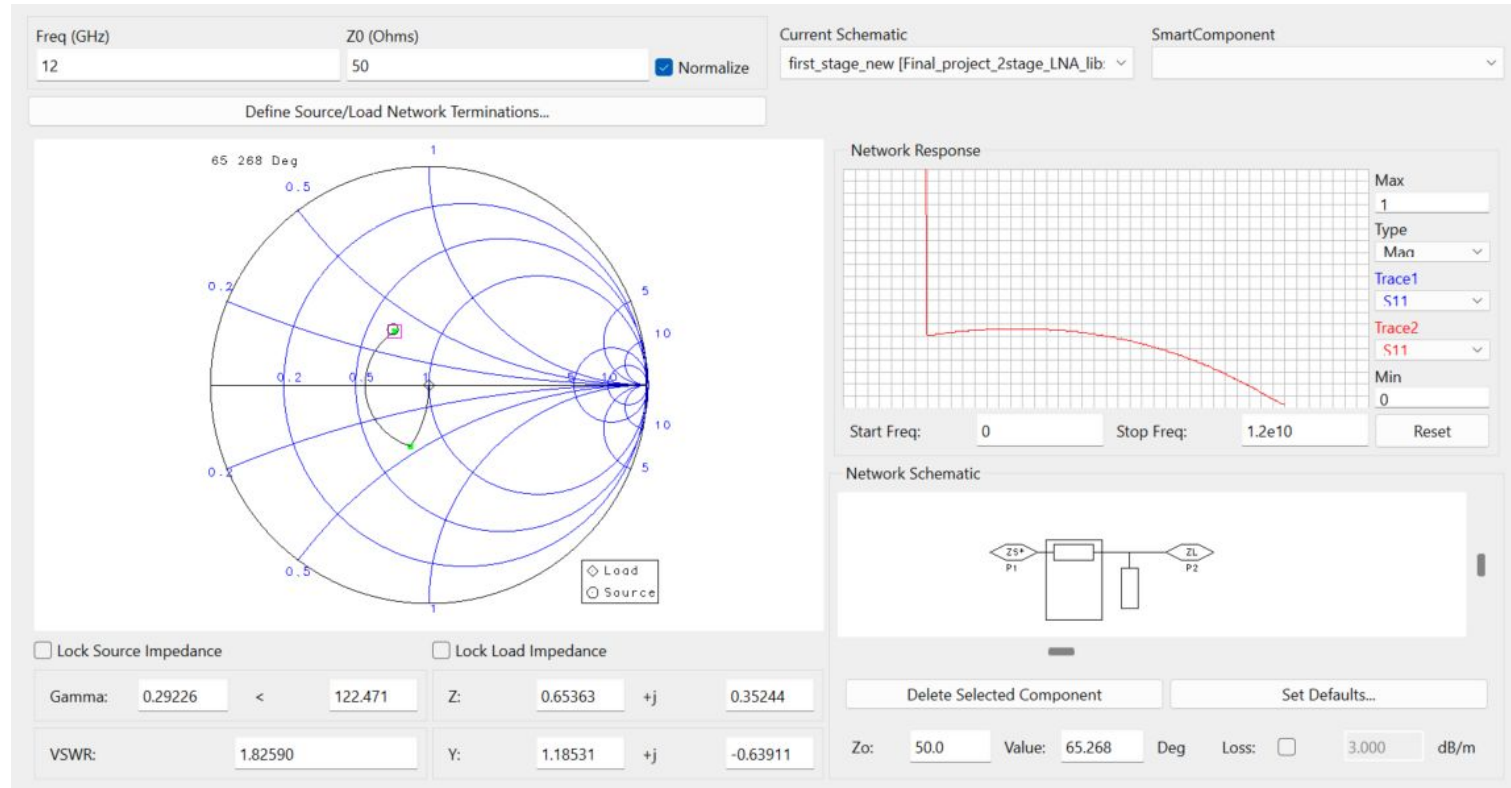
m2
freq=12.00 GHz
NFmin=0.438



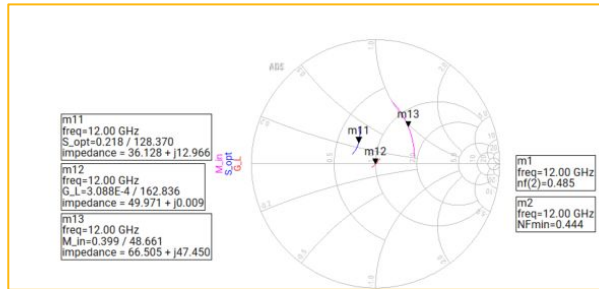
- I first matched the Sopt between the GL and 50 ohm.
 - **This was a mistake as GL is the conjugate of the Gout not the conjugate of the GS
- I was hoping to get more gain out of the first stage by doing this.



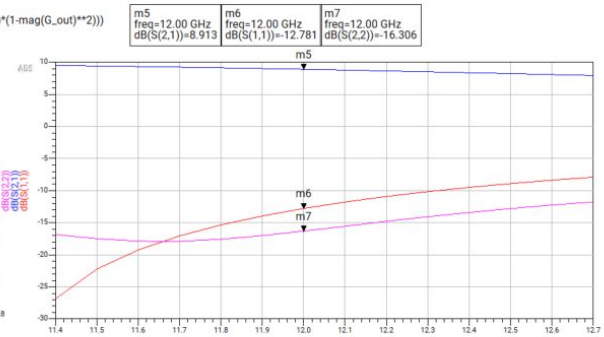
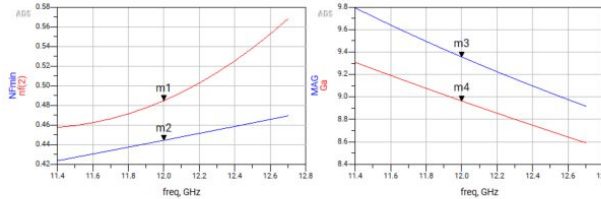
GL Match



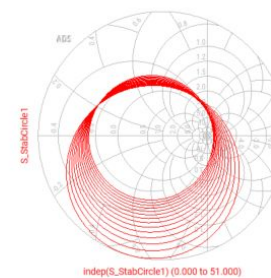
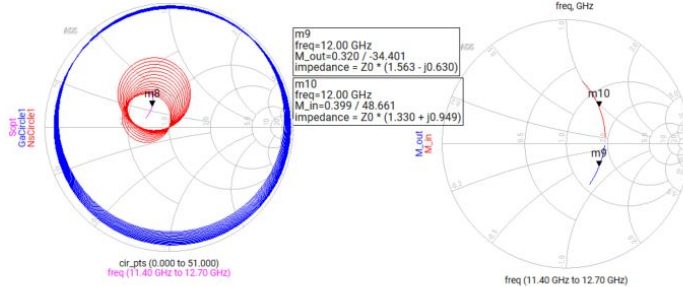
First Stage with Sopt match and GL match



$$\begin{aligned}
 \text{EqnM_in_sm_gamma1(S)} & \\
 \text{EqnM_out_sm_gamma2(S)} & \\
 \text{EqnK=stab_fact(S)} & \\
 \text{EqnMAG=max_gain(S)} & \\
 \text{Eqndelta=mag(S11*S22*S12*S21)} & \\
 \text{EqnGa} &= 10*\log((\text{mag}(S21)**2)*(1-\text{mag}(Sopt)**2)/((\text{mag}(1-S11*Sopt)**2)*(1-\text{mag}(G_out)**2))) \\
 \text{EqnG_out} &= (S22+S12*S21*Sopt/(1-S11*Sopt)) \\
 \text{EqnG_L} &= \text{conj}(G_out) \\
 \text{EqnS_opt} &= Sopt
 \end{aligned}$$



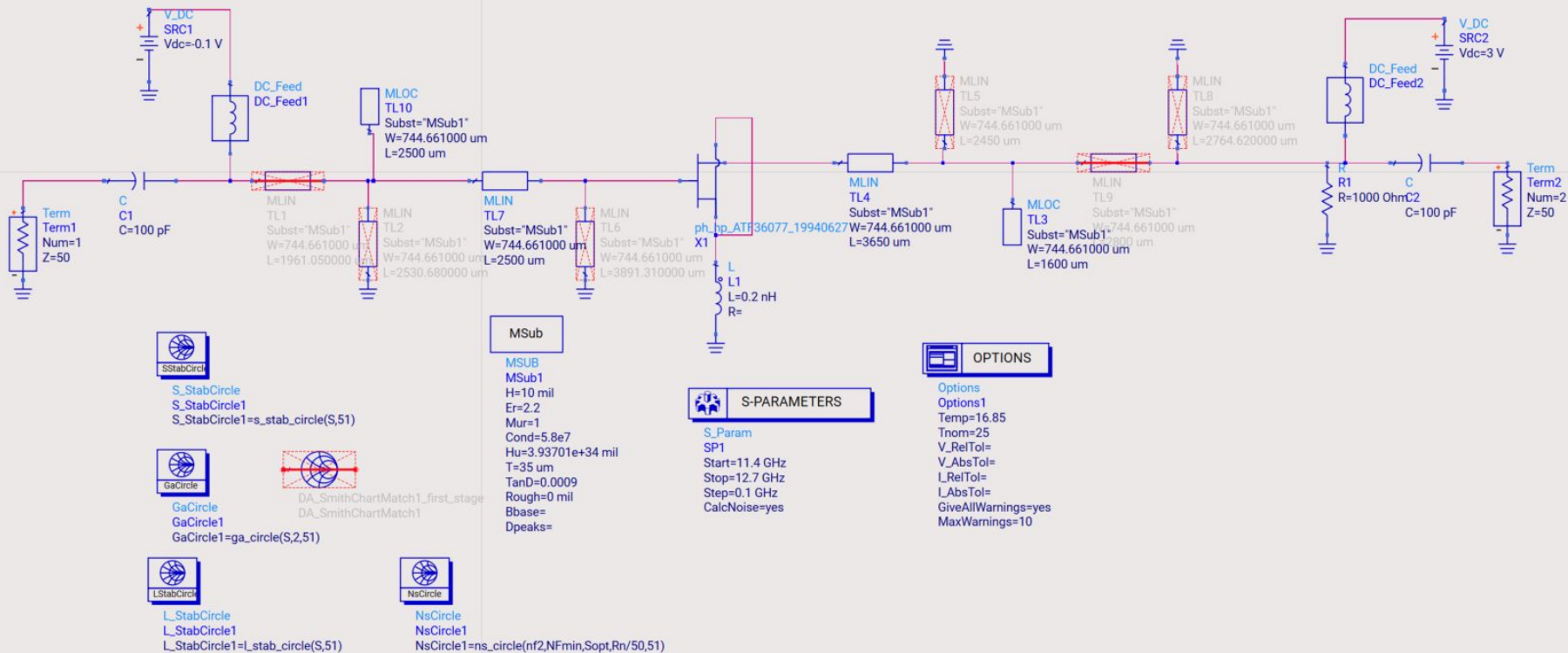
You can see in this slide the first stage has good noise figure, decent gain, and decent S-parameters. The matching is also where we want it.



freq	k	delta	G_L	S_opt	freq
11.40 GHz	1.033	0.722	0.040 / -137.908	0.204 / 157.653	11.40 GHz
11.50 GHz	1.034	0.718	0.034 / -137.422	0.199 / 152.741	11.50 GHz
11.60 GHz	1.035	0.715	0.027 / -136.974	0.197 / 147.548	11.60 GHz
11.70 GHz	1.037	0.712	0.020 / -136.601	0.196 / 142.300	11.70 GHz
11.80 GHz	1.038	0.709	0.014 / -136.416	0.202 / 137.231	11.80 GHz
11.90 GHz	1.039	0.706	0.007 / -136.969	0.209 / 132.542	11.90 GHz
12.00 GHz	1.040	0.703	3.088E-4 / 162.836	0.218 / 128.370	12.00 GHz
12.10 GHz	1.041	0.700	0.007 / 48.771	0.229 / 124.786	12.10 GHz
12.20 GHz	1.042	0.697	0.014 / 48.190	0.243 / 121.798	12.20 GHz
12.30 GHz	1.042	0.695	0.021 / 48.327	0.257 / 119.375	12.30 GHz
12.40 GHz	1.043	0.692	0.028 / 48.702	0.273 / 117.467	12.40 GHz
12.50 GHz	1.044	0.689	0.035 / 49.157	0.289 / 116.011	12.50 GHz
12.60 GHz	1.045	0.687	0.042 / 49.653	0.306 / 114.947	12.60 GHz
12.70 GHz	1.045	0.684	0.049 / 50.173	0.324 / 114.216	12.70 GHz



First Stage Schematic - First Pass

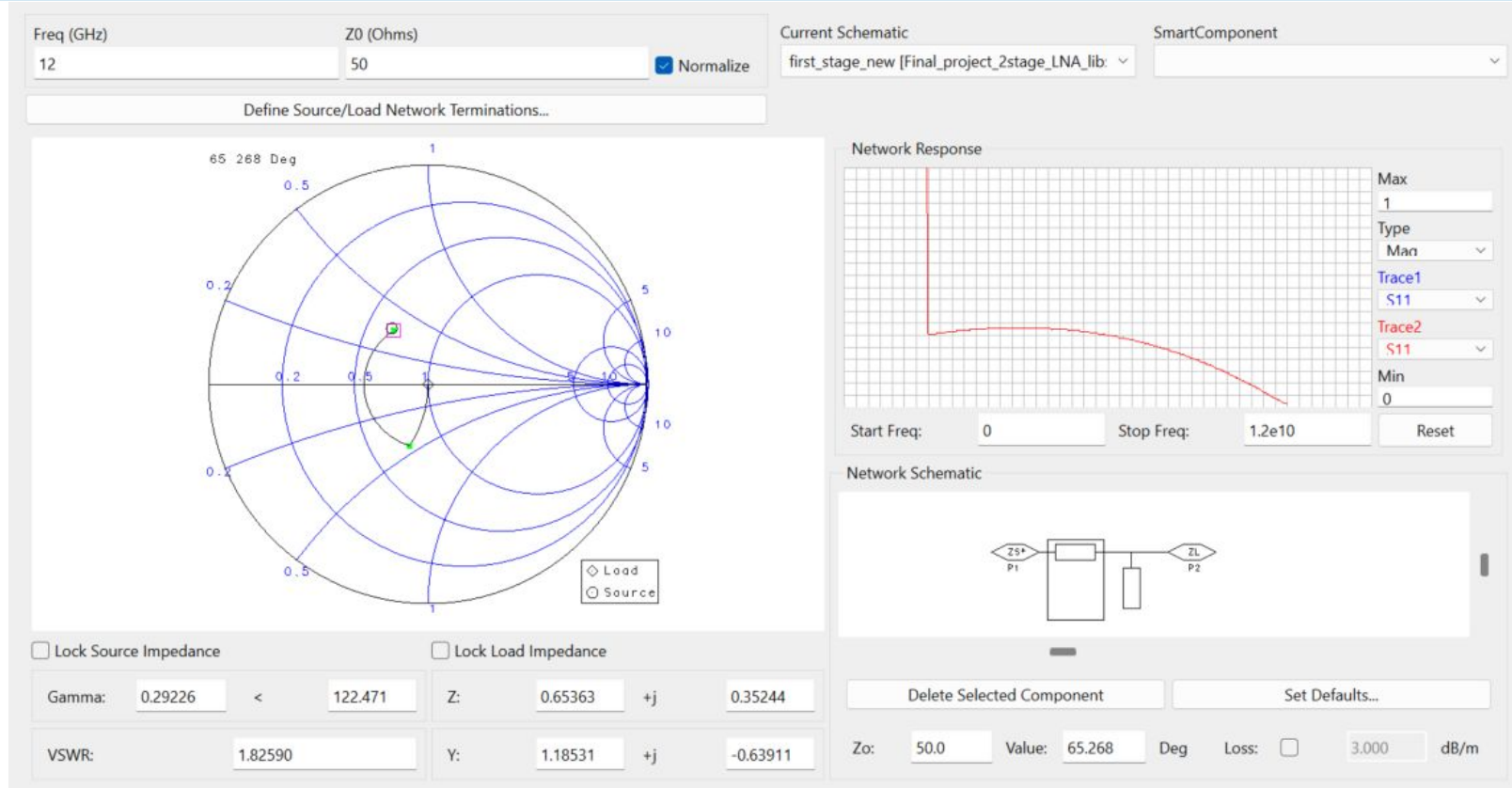


Second Stage

Optimize for Gain

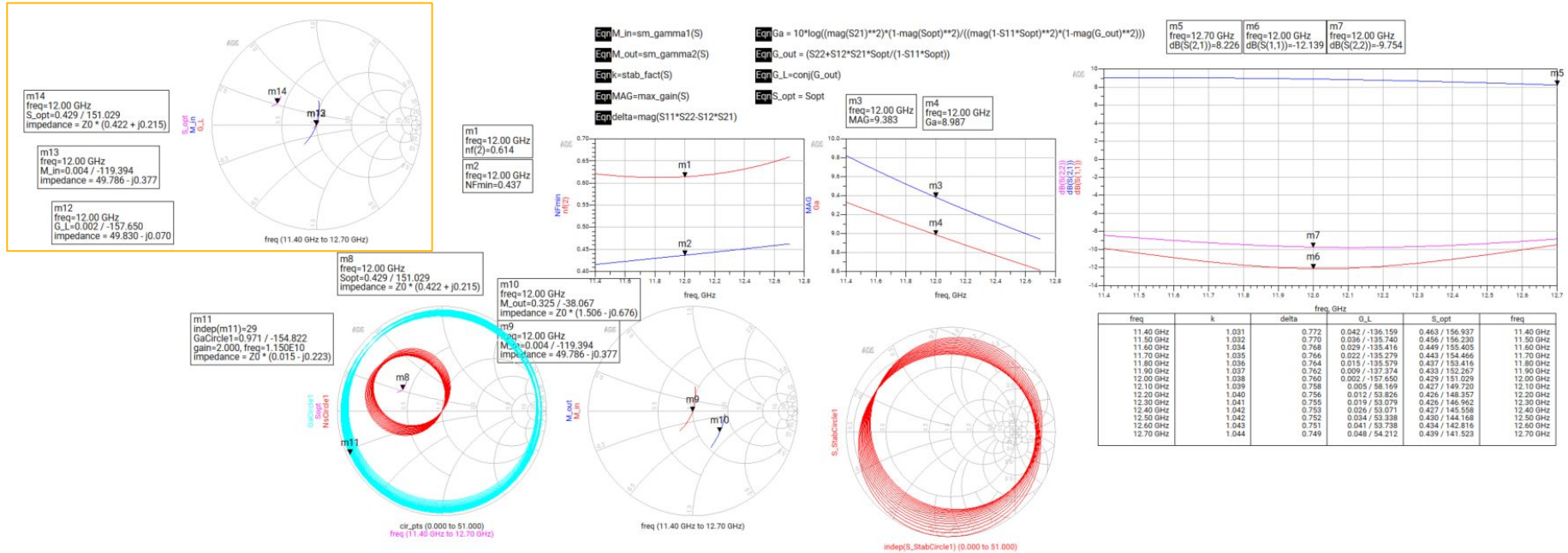


Conjugate matching for the output

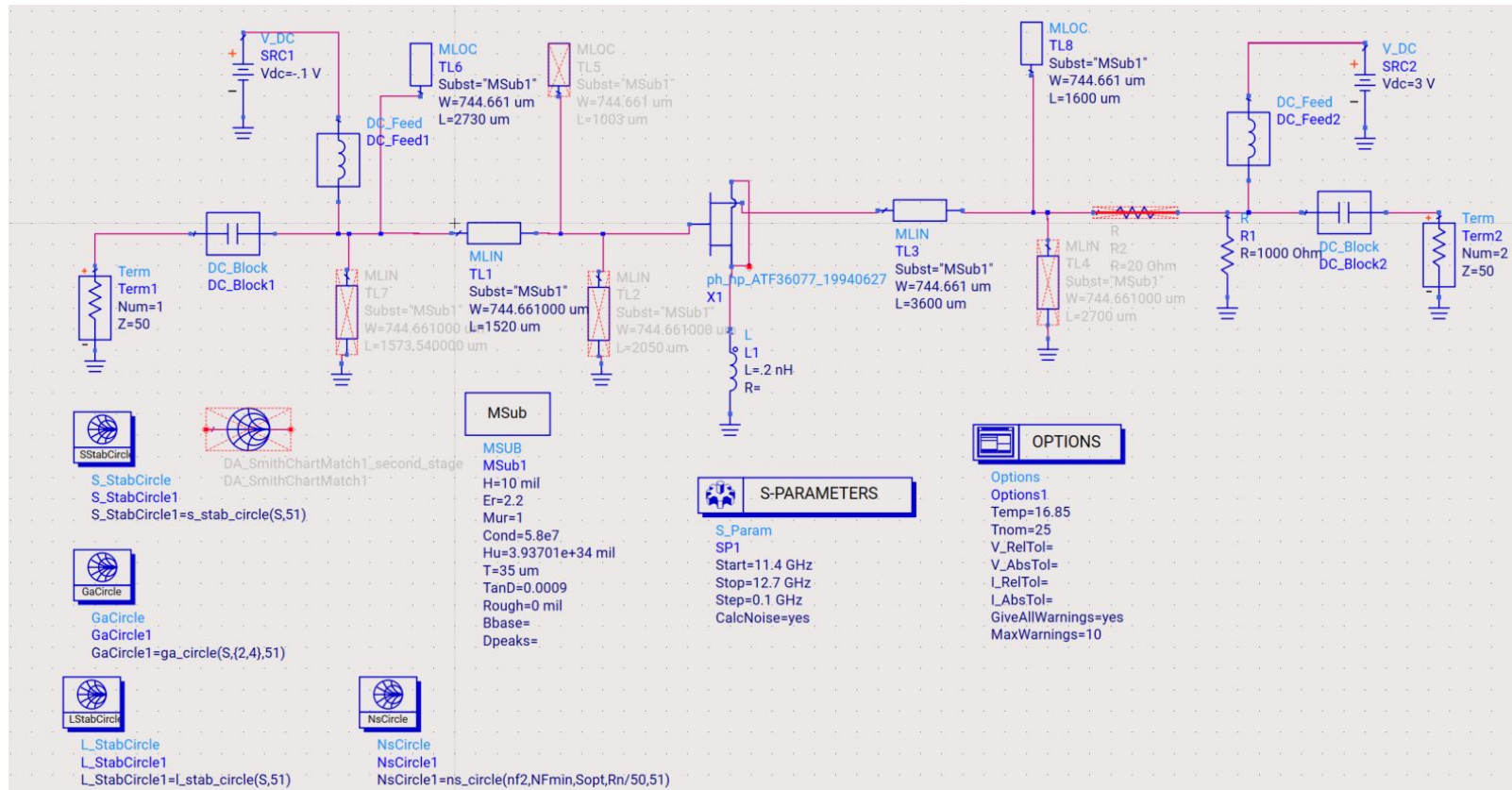


Second Stage - Matched output

- I first conjugate matched the output to 50 Ohms. This can be seen in the highlighted box on the left side.
- Next match the input to 50 Ohms, which can also be seen with the Min at 50 Ohms.



Second Stage Schematic - First Pass

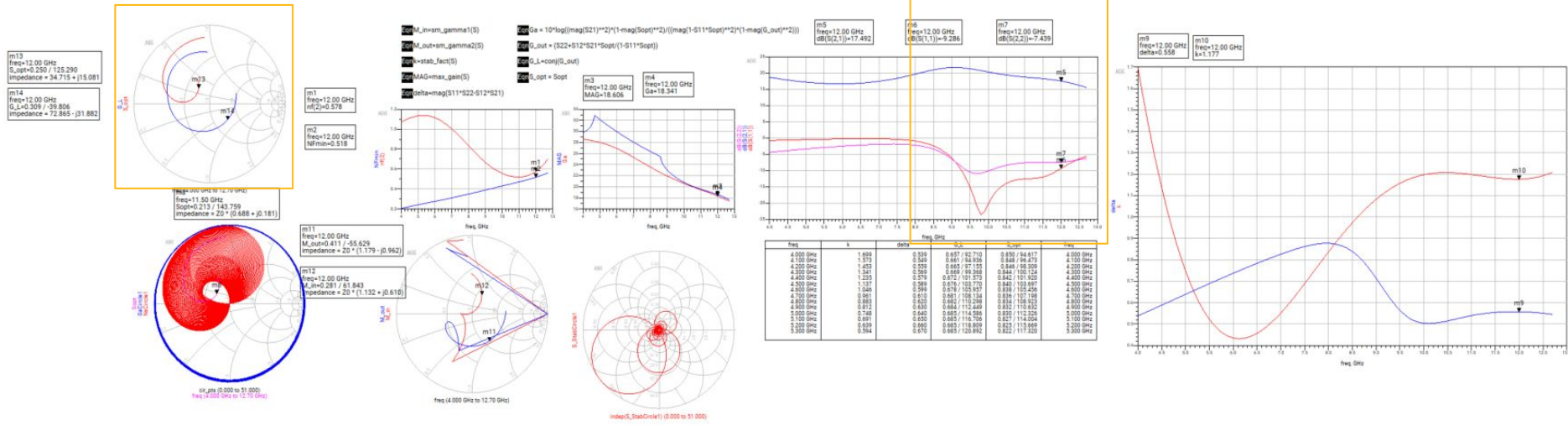


Integration



Integrating both stages

- The Sopt seems to be steady, but the GL drifted. This is causing the metrics to be off from what we calculated.
- I then rematched the GL to the 50 Ohms and this gave better results (shown in the next slide), but still not good enough.



First Pass - results

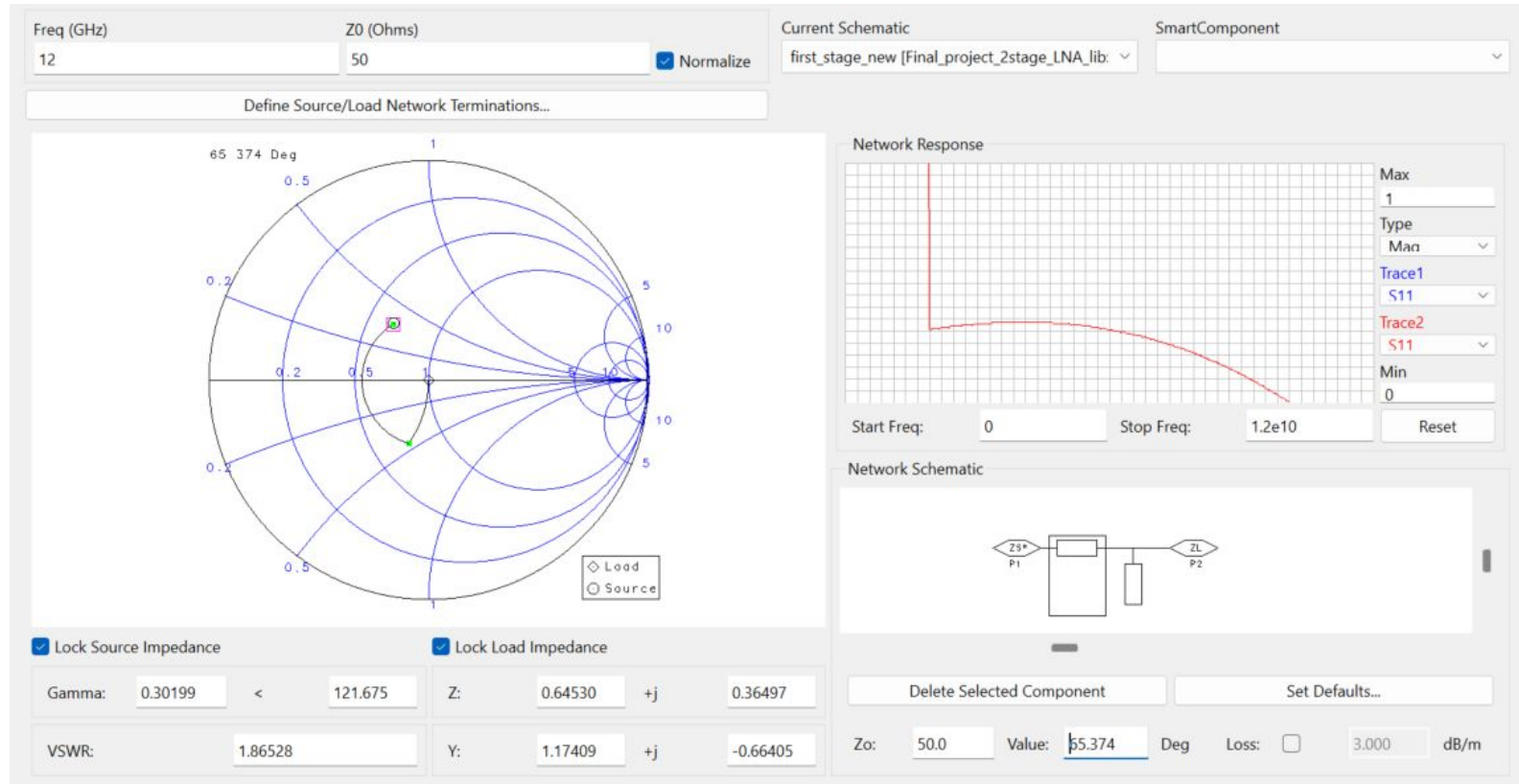
Characteristic	Specification
Operating Frequency	12 GHz
Noise Figure	.577dB
Insertion Gain (S21)	20.2dB
Output Reflection (S22)	-16.4dB
Input Reflection (S11)	-12.8dB
Stability	Unstable from 4.5GHz-9GHz

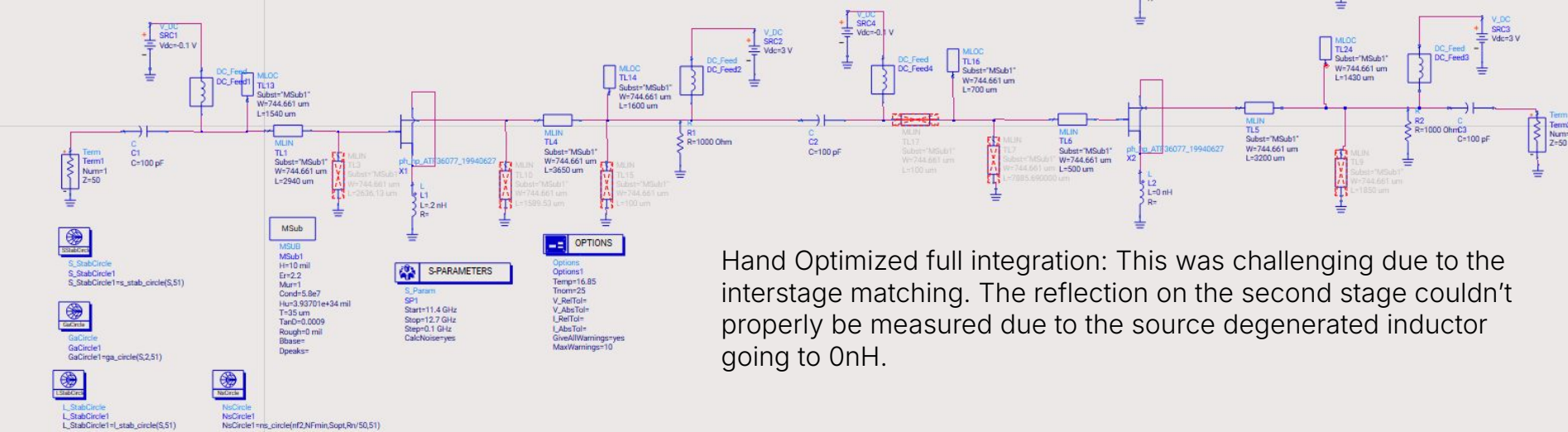
- At a first glance this looks great, but this is only at the designed frequency.
- You can clearly see the gain and S-parameters drift out of spec near the 12.5GHz region.
- The instability is attributed to the transistor model and so this might be clear

Hand Optimization

Much closer matching to the 50 ohm

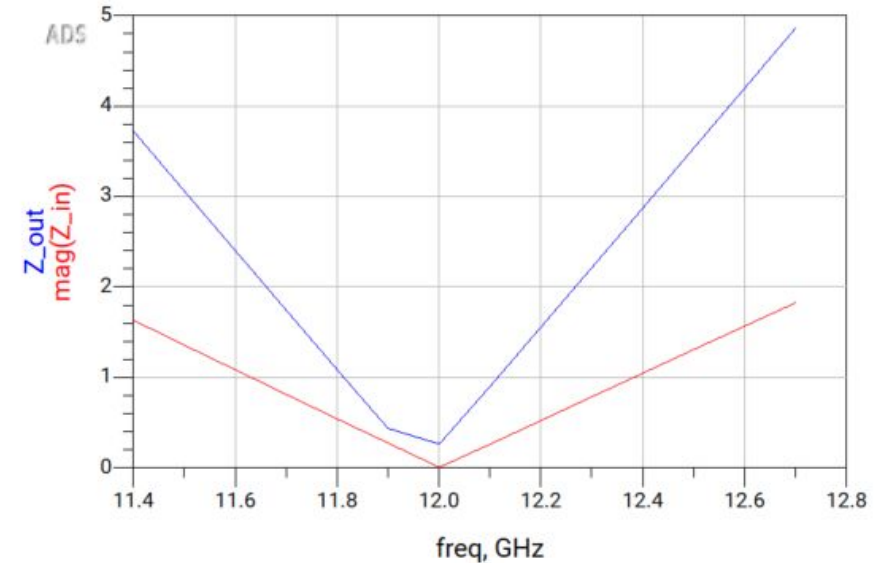
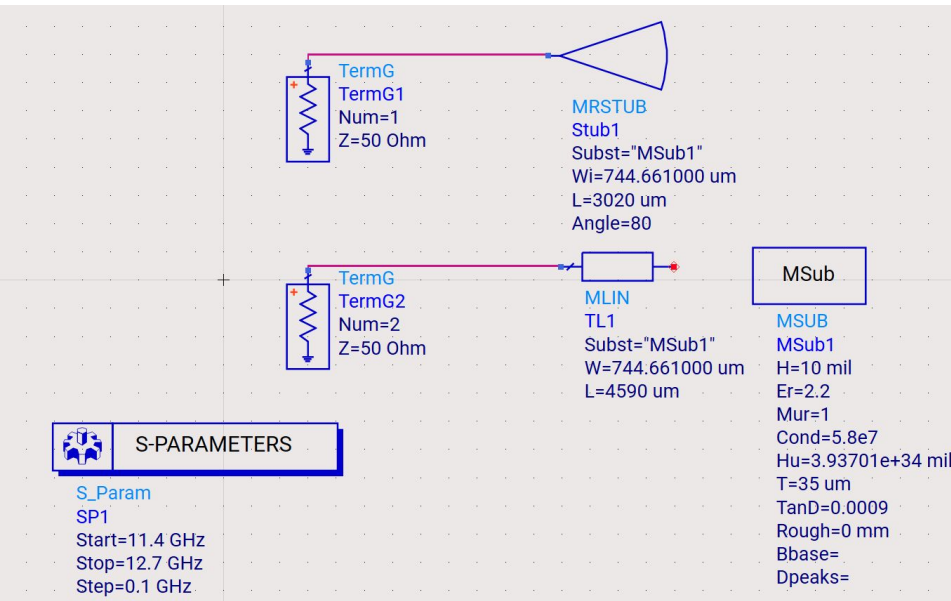
Fixing the Match to Sopt on the first stage





Biasing

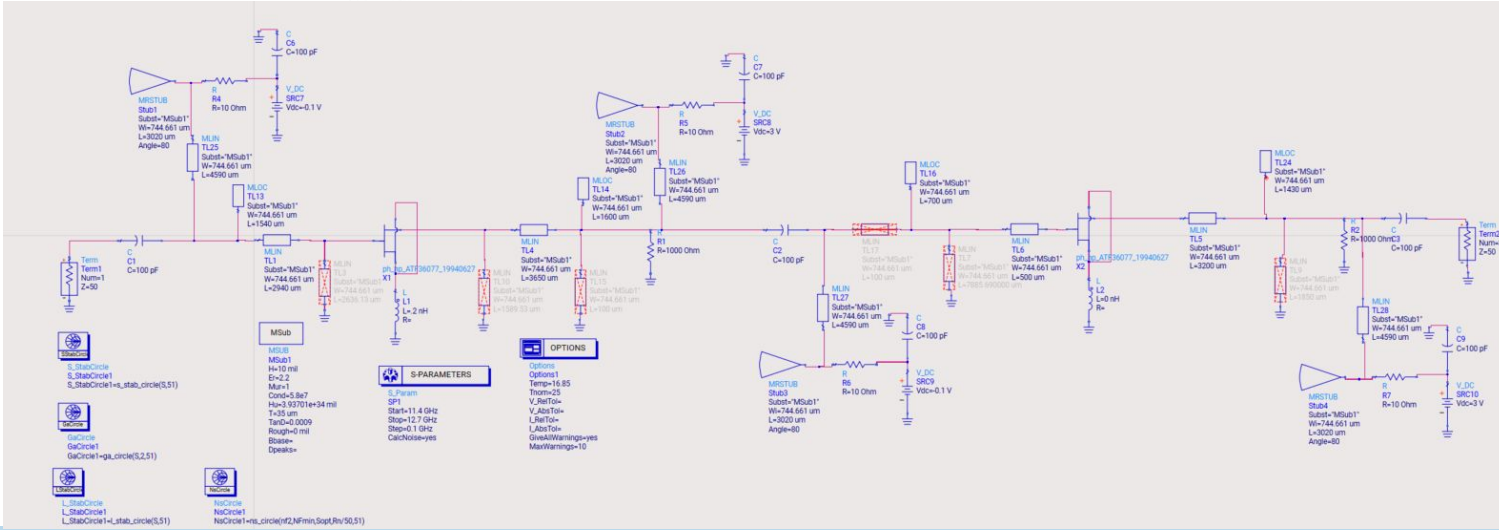
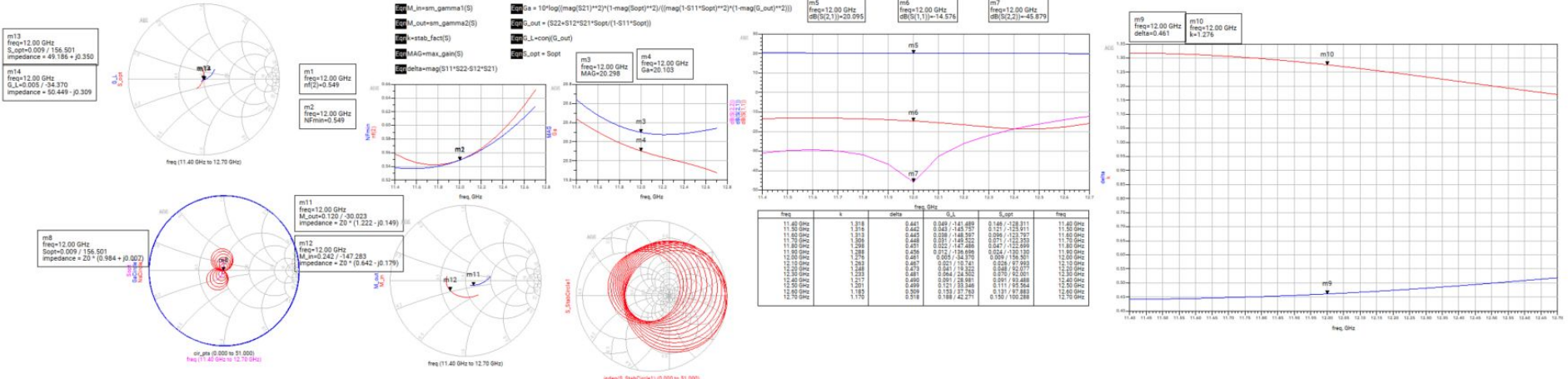
- Replacing Open Stubs with Radial Open Stubs for higher bandwidth short
- Replacing bias networks with a quarter-wavelength line, resistor, and capacitor, integrating into matching network where possible



$$\text{Eqn } Z_{in} = z_{in}(S(1,1))$$

$$\text{Eqn } Z_{out} = 50 * (1 + S(2,2)) / (1 - S(2,2))$$



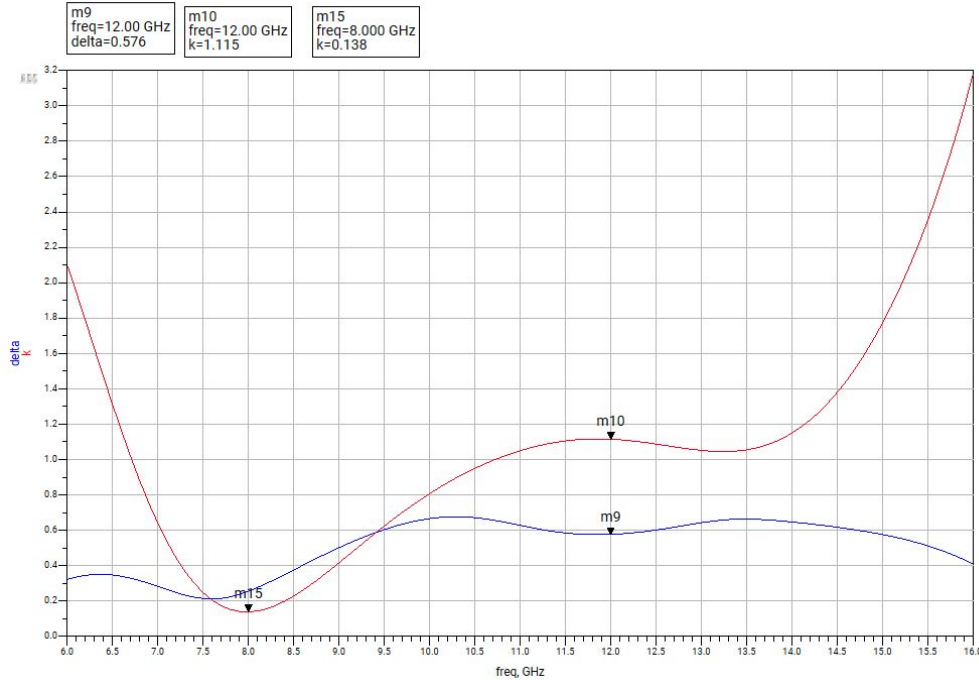


ADS Optimization

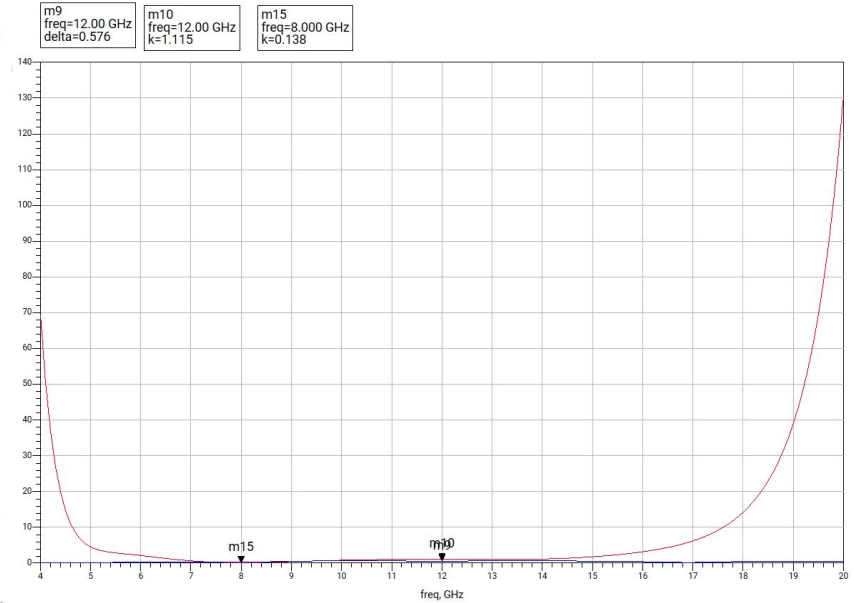


Unconditionally Stable

Zoomed in to show waveform.



Full range to demonstrate stability. (cutoff at 4-20GHz for viewing purposes. At DC k runs too high to see the rest of the plot.)

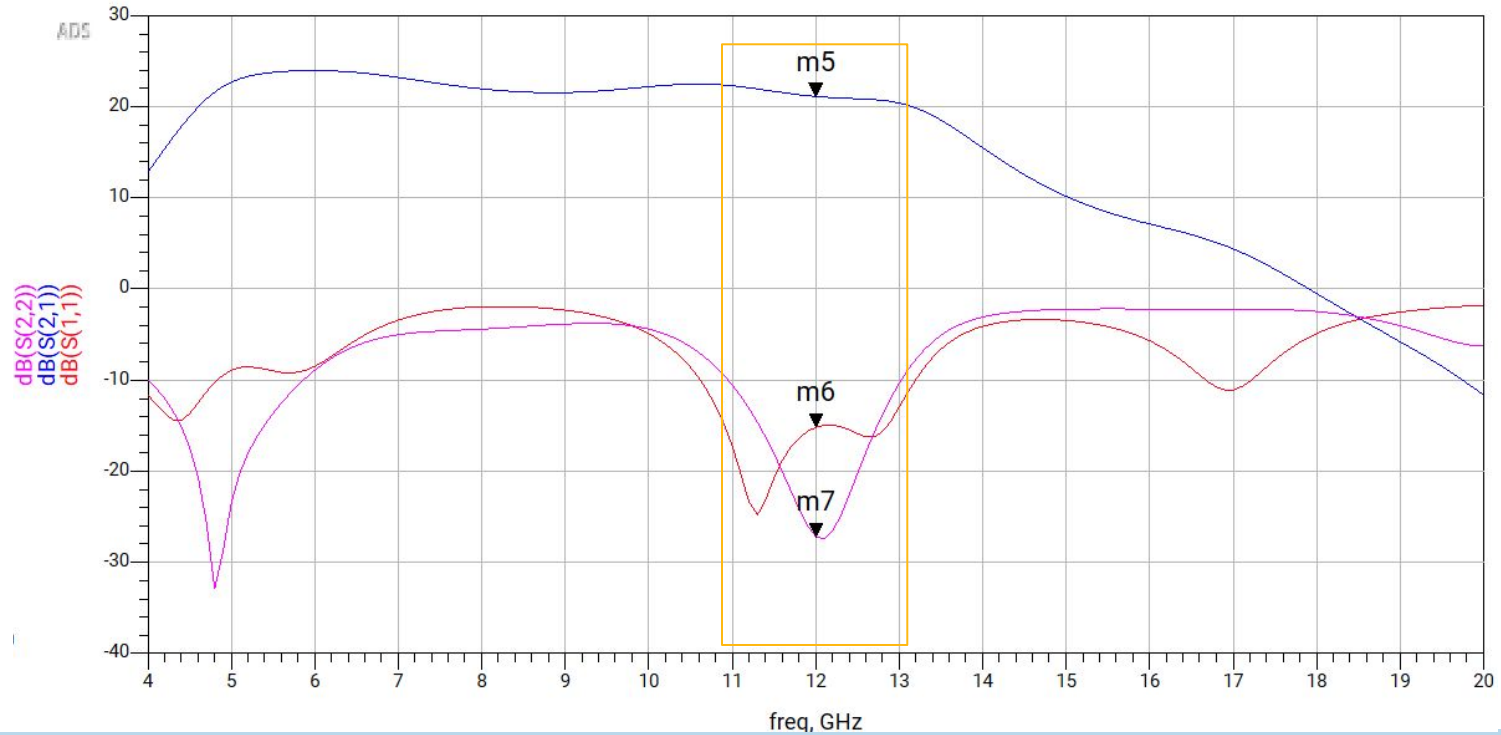


S-Parameters

m5
freq=12.00 GHz
dB(S(2,1))=21.112

m6
freq=12.00 GHz
dB(S(1,1))=-15.176

m7
freq=12.00 GHz
dB(S(2,2))=-27.217



Final - results

Characteristic	Specification @ 12 GHz
Operating Frequency	11.4-12.7 GHz
Noise Figure	.534<1dB
Insertion Gain (S21)	21.112>20dB
Output Reflection (S22)	-27.217<-14dB
Input Reflection (S11)	-15.175<-10dB
Stability	Unconditionally Stable 0.1-20GHz

- Cleared all requirements through the entire bandwidth

DC Power

Power from Source 1:

- $P = V \cdot I = (0.1)(17.1 \text{ fA}) = 0 \text{ W}$

Power from Source 2:

- $P = V \cdot I = (3)(22.5\text{mA}) = 67.5\text{mW}$

Power from Source 3:

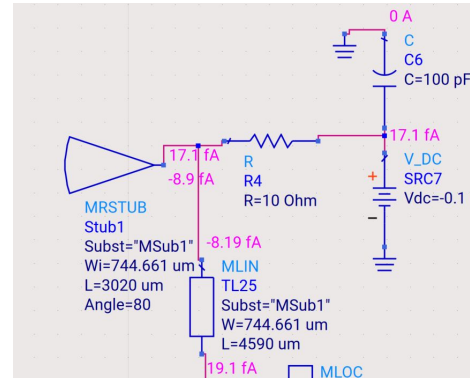
- $P = V \cdot I = (0.1)(21.4 \text{ fA}) = 0 \text{ W}$

Power from Source 4:

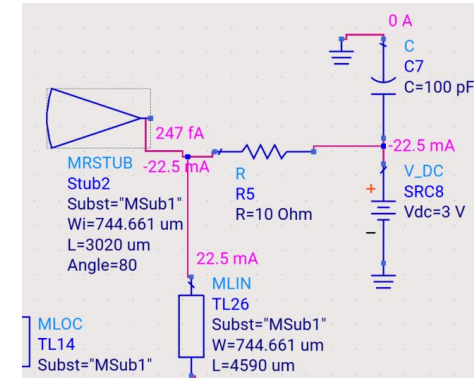
- $P = V \cdot I = (3)(22.5\text{mA}) = 67.5\text{mW}$

Total Power from Sources: 135mW

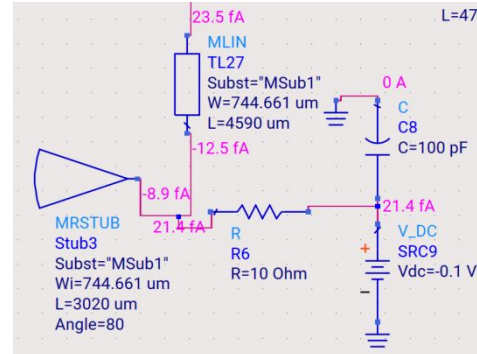
Source 1



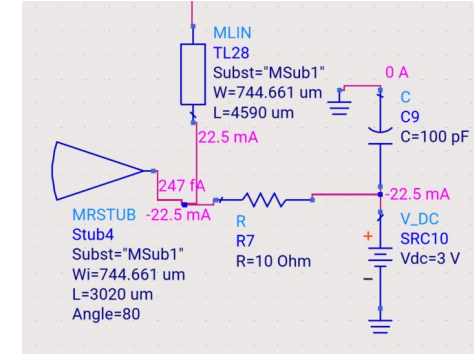
Source 2



Source 3



Source 4



DC Power Across Resistive Loads

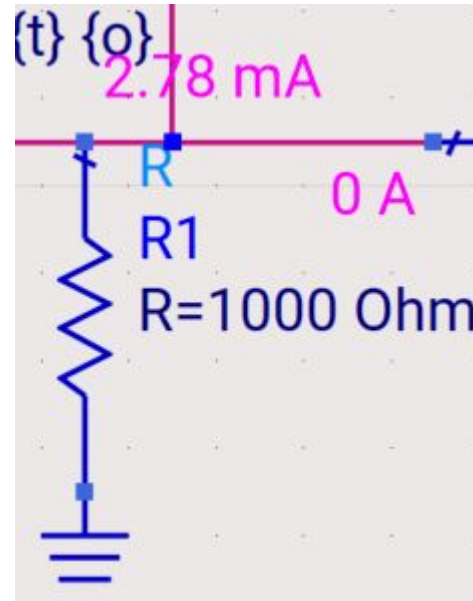
Power Across 1:

- $P = I^2 * R = (1K)(2.78mA)^2 = 7.7284mW$

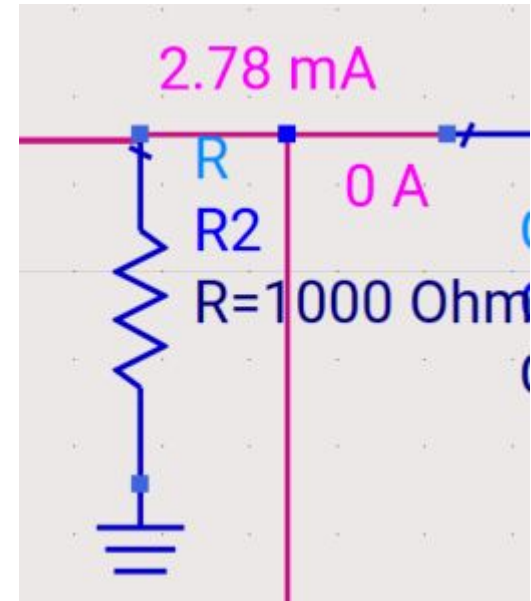
Power Across 2:

- $P = I^2 * R = (1K)(2.78mA)^2 = 7.7284mW$

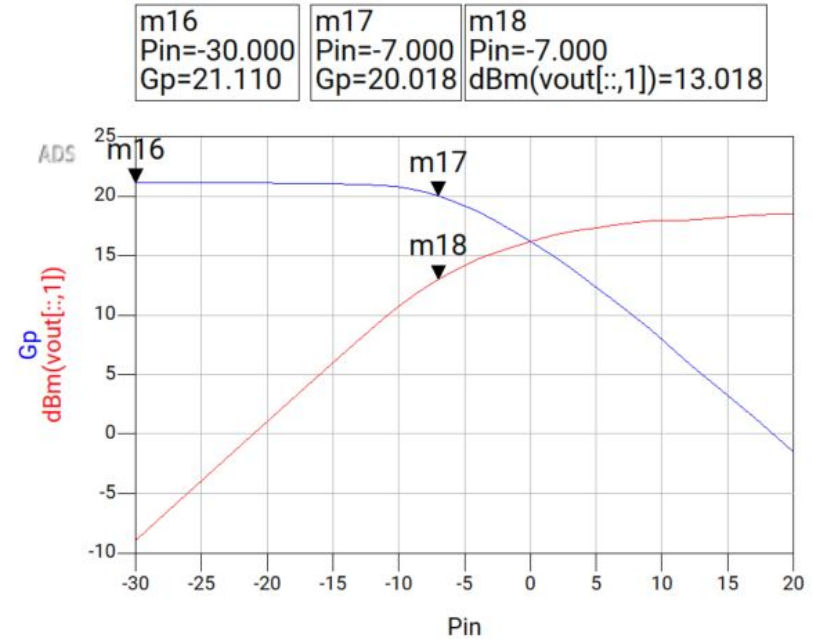
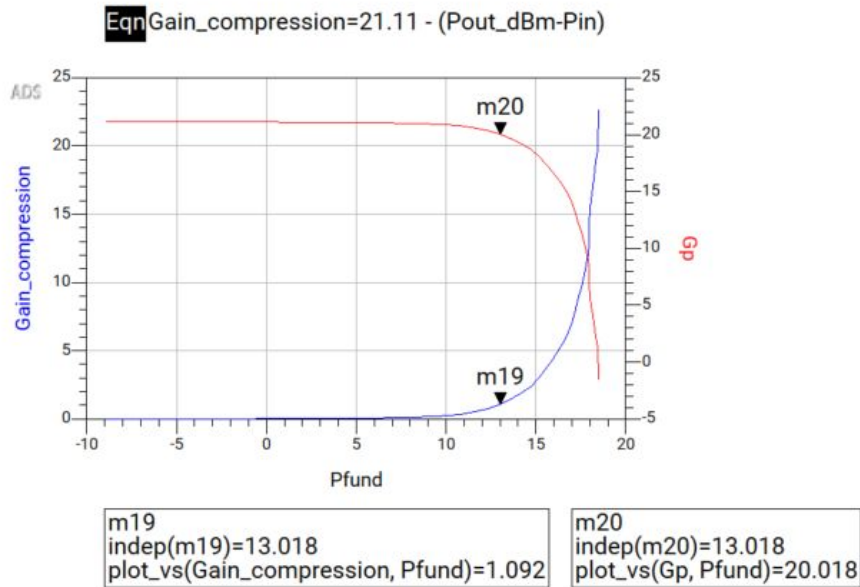
Resistive Load 1



Resistive Load 2

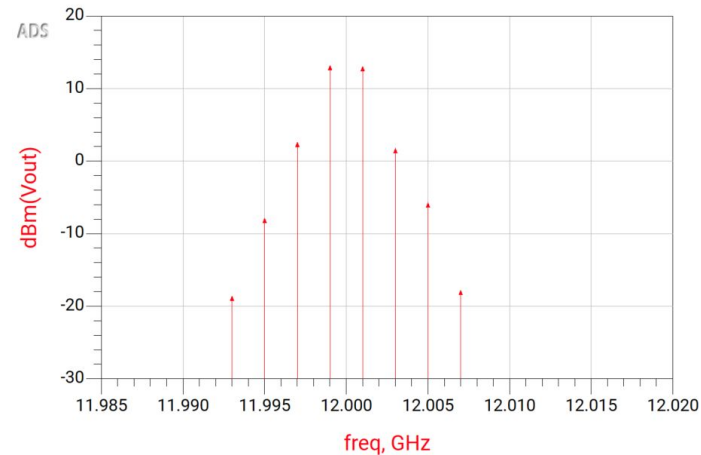
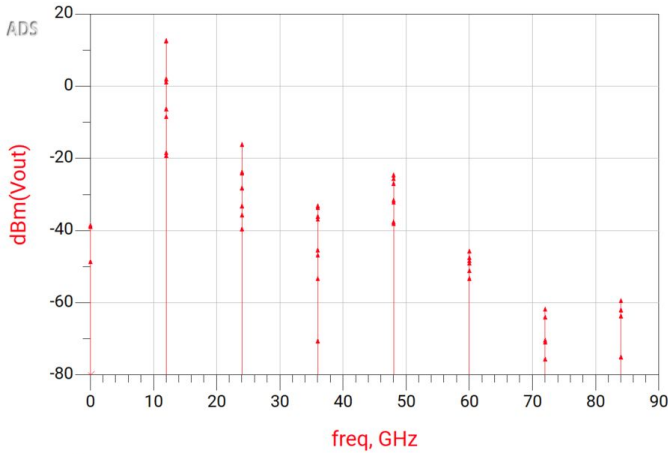


P1dB Compression



P1dB Input = -7 dBm
P1dB Output = 13 dBm

IIP3/OIP3 - Two Tone Harmonic Balance



IP3inLow	IP3inHigh	ipoHigh	ipoLow
18.515	18.688	18.688	18.515



Thank You!!